Examiner reports

Q1.

This question proved quite challenging with around 40% of students choosing the correct second option, slightly ahead of the 35% who chose the third option.

Q2.

Here the similarity of the answers resulted in many students making the wrong choice. The correct answer was most popular, but was only chosen by 35%. The second favourite was option 1, suggesting that many calculated the gradient of the radius from the origin to the centre rather than the tangent at the origin.

Q3.

Here it was insufficient to state the midpoint or the gradient of the perpendicular bisector. These facts needed to be applied in a method leading towards finding the values of c and d.

Many correctly obtained the required gradient equation. Fewer substituted the coordinates of the midpoint into the equation of the bisector. Many instead stated that *C* and *D* lay on the bisector leading to wrong equations. Even where the two correct equations were obtained, poor algebra often led to wrong answers. A common error was to say that since $\frac{d-2}{6-c} = \frac{1}{4}$ then $d = 3$ and $c = 2$.

Q4.

This question illustrated that many students were unclear of the geometry required to prove that the given four points formed a rectangle. Many students scored 2/4 for showing that the points formed a parallelogram, but investigating for right angles was less commonly seen. Some students incorrectly assumed that the distances between the points were vertical and horizontal. The final mark required a rigorous statement which proved beyond most students. There were many different acceptable methods of proving that the points formed a rectangle.

There were many correct attempts seen for part (b) with 60% of all students obtaining full marks.

Q5.

In part (a) students had a choice of techniques to use and the first mark was awarded for using an appropriate technique to solve the problem. This was a “show that” question, which was an indication that a rigorous argument needed to be given. Those who compared gradients often failed to complete their argument with a statement such as “$m_1 \times m_2 = -1$ means that the lines are perpendicular, so *ABC* is a right angle.”

Part (b)(i) was intended to help students to answer part (b)(ii). While most students seemed to know the correct theorem, it was usually poorly stated.

Part (b)(ii) was generally done well, with students making good progress. The main errors seen here were numerical slips or failing to give a justification for their answer. Many students showed clear working, gave an argument to show the distance from the centre to *D* was greater than the radius and concluded that *D* lay outside the circle.

Q6.

This was the least well answered of all the multiple choice question on the paper. However, a large majority of students still gave the correct answer of $3x - 2y = 7$.

Q13.

(a) (i) Those who began by substituting $x = 7$ into the equation of AB were usually successful. Another approach was to rearrange the given equation in order to find the gradient and then equate this value to the gradient found using the points (–3, 2) and (7, k). Quite a few attempted to verify the result by more complicated methods, but such attempts rarely had an appropriate conclusion and did not always convince the examiner.

(ii) Apart from the occasional arithmetic error, this was answered well. A few weaker students used an incorrect midpoint formula, usually having minus signs instead of plus signs.

(b) Once more this was answered well. The most successful students rearranged the given equation to make y the subject, but others used the two pairs of coordinates. A few students made sign errors due to the negative coordinates but most students found the correct value of the gradient.

(c) Most students realised the need to find the negative reciprocal of the gradient of AB in order to find the gradient of the perpendicular line. Very few failed to do so. However, not all heeded the question where a specific form of answer was requested. A common incorrect answer was $5x - 3y + 7 = 0$ where students, having obtained the equation $y = \frac{5}{3}x + 7$, neglected to multiply the constant term by 3. Those who used $y = mx + c$ for the equation of the straight line usually made more mistakes than those using the form $y - y_1 = m(x - x_1)$.

(d) There were several completely correct solutions here. The question was easier than on previous papers because students did not have to decide which equations to use to find the coordinates of C. However, amongst the weaker students, sign and arithmetic errors abounded.

Q14.

This question was not done well, with fewer than 10% of the entry gaining full marks, although over half did gain 4 or more of the 8 marks available.

Part (a)(i). Most students were successful here; using implicit differentiation as expected and clearly showing the derivatives of the three terms. Those few who solved y in terms of x before differentiating were rarely successful, particularly those who thought that $y = x - \sqrt{8}$. Some students left their derivative in terms of x and y rather than the requested p and q.

Part (a)(ii). Many students gained the first 2 marks for a correct expression for the tangents, especially those who used $y - y_1 = m(x - x_1)$. The relatively few who tried to use $y = mx + c$ often made an error in finding c in terms of p and q. The common mistake was a sign error in the gradient at $(p, -q)$, reducing the attempt at solving the two tangent equations simultaneously to nonsense. The simplest way to establish the requested result is to add the two tangent equations, but few students did this. Most equated the two expressions for y and attempted to find x, often making an algebraic error. This is a valid method if the expression for x is substituted back in a tangent equation and $y = 0$ is shown, but few attempted to do this. Other students assumed $y = 0$ and attempted to find an expression for xbut most were unsure how to proceed from there. Other students wrote their tangent equations in the form $ax + by + c = 0$ and equated them both to zero. Again it is possible to solve the problem this way, although most students could not complete successfully.

Part (b). A variety of attempts at this question were seen, including those who thought they had been asked to find the derivative using the parametric equations, which were irrelevant. Some students seemed not to understand the question at all. Those who squared the expressions for x and x and subtracted were usually successful, although many made an error in the squaring by missing a term or making a sign error. Very few factorised $x^2 - y^2$ and then used the parametric equations to eliminate t, but some students did establish the given result by other algebraically valid eliminations of t.

Q15.

(a) (i) Almost all candidates began by substituting $x = p$ and $y = p + 2$ into the given equation. However, many made a sign error when multiplying out $-4(p + 2)$ and consequently $p = 13$ was a very common incorrect answer.

(b) Most candidates rearranged the equation to the form $y = mx + c$ and gave the correct gradient. Some used the coordinates of two points on AB, such as (1, 2) and (–3, –1). Some of the weaker candidates chose two values for p, found 2 pairs of incorrect coordinates and then attempted to find the gradient of the line joining these points.

(c) There were two equally common methods used here. Both required the use of the negative reciprocal of the answer to part (b) to provide the relevant gradient and it was pleasing to note that most candidates realised this. One approach was to find the gradient in terms of k and equate it to the relevant gradient and solve; the alternative was to set up an equation using the coordinates of A (or C) and then to substitute the coordinates of the other point to obtain the value of k.

(d) Although most candidates realised the necessity to set up 2 equations in order to eliminate x or y, arithmetic and sign errors abounded and there were many incorrect solutions. Even those who correctly found $7y = -28$, say, often wrote down an incorrect value of y such as $y = -3$ or $y = 4$. A few candidates misread the question and used an incorrect equation for one of the lines.

Q16.

(a) Those candidates who began by finding $\frac{dy}{dx}$ usually found the correct gradient and most gave the equation of the tangent in the required form. A few made arithmetic errors such as $y - 6 = 9x + 9$ leading to $y = 9x + 3$. A few, having found the gradient to be 9, then incorrectly used the negative reciprocal , i.e. $-\frac{1}{9}$. Some candidates did not realise that differentiation was required and attempted to find the gradient using the coordinates of the two points (–1, 6) and (2, 33).

(b) (i) Provided no arithmetic errors were made, particularly evaluating $2^5 = 64$, then the correct value of k was found from the equation of the curve. A few candidates used the tangent equation and although they found that $k = 33$, no credit was given.

(ii) Those who used the equation of the line did not always earn the mark as a concluding statement such as "therefore lies on the tangent" was not made. Some worked backwards and found the gradient using (2, 33). A few candidates confused the requests in parts (i) and (ii) and consequently earned no marks.

(c) (i) The majority of candidates integrated correctly. Most substituted the limits in the correct order but many sign and arithmetic errors were made. A common error was evaluating $\frac{5}{6} - \frac{54}{6}$ as $\frac{49}{6}$ instead of $-\frac{49}{6}$ and the correct value of the integral was rarely seen.

(ii) Very few candidates found the area of a trapezium. Most found the area of a triangle instead. Those who used integration in order to find the area under the straight line were far more successful than those who tried to remember the basic formula from GCSE.

Q17.

In part (a), almost everyone recognised the need to use the distance formula, though a few were unable to calculate 49 + 4 successfully. Many candidates scored full marks for correct working and a correct summary statement. Those who failed to write a correct conclusion involving the lengths OA and OB lost the final mark. Several candidates confused OA^2 with OA , writing things like $OA^2 = 6^2 + (-4)^2 = \sqrt{52}$, which again was enough to lose the final mark. A minority of candidates used an incorrect formula for the distance between two points, such as $OA = \sqrt{36 - 16} = \sqrt{20}$ and scored no marks at all.

In part (b)(i), apart from a few sign errors, most candidates obtained the correct gradient. Some wrote the gradient as "difference in x values / difference in y values" and so did not score any marks.

In part (b)(ii), most candidates wrote down a correct form of the equation, such as $y - -4 = \frac{-11}{8}(x - 6)$. However, careless arithmetic prevented many from obtaining the final equation in the required form with integer coefficients. Those who used $y = mx + c$ often only earned a single mark for having the correct gradient, as they never had a correct form of the equation.

In part (c), most candidates realised that the product of the gradients of perpendicular lines should be –1 and wrote the perpendicular gradient as $\frac{8}{11}$. Many made little progress beyond this point; some used the coordinates of B instead of A ; others used the correct method but combined 44 and 48 incorrectly to give 82, or even – 4. Full marks were given for giving an answer as a fraction not reduced to its simplest form.

Q18.

In part (a), most candidates were able to complete the square for at least one of the terms. A very common incorrect answer involved $(x - 7)^2$ instead of $(x + 7)^2$. Occasionally a negative answer was given for the square of the radius. Also, $\sqrt{5}$, $\sqrt{25}$ and ± 49 were often seen on the right-hand side of the equation.

In part (b), the coordinates of the centre usually followed through correctly from part (a). Often the correct radius was stated, even though part (a) was incorrect.

In part (c), the centre of the circle usually appeared in the correct quadrant for the candidate's coordinates. However, those who had the correct answers in part (b) did not always draw the correct circle touching the x -axis at –7 and not touching the y -axis.

In part (d)(i), several candidates earned the method mark but the most common error lay in the failure to square (kx) as k^2x^2; most wrote kx^2 and immediately lost the accuracy mark.

In part (d)(ii), candidates seemed to think it necessary to write "= 0" only on the very last line. This does not fulfil the requirements of a proof. The first line should be that equal roots occur when $b^2 - 4ac = 0$ and the "= 0" should continue throughout.

Other common errors occurred in the squaring of $2(k + 7)$, which often became $2(k + 7)^2$. Candidates should check each line of their proof instead of fudging one or more lines of working, as examiners are unlikely to be deceived by the miraculous appearance of the printed answer following incorrect working.

Many did not attempt part (d)(iii). The factorisation of the quadratic expression defeated most candidates; too many are reliant on the quadratic equation formula which often produces large numbers that are difficult to handle without a calculator. The incorrect factors $(4k - 3)(3k + 4)$ were seen as often as the correct ones. Candidates found this factorisation extremely hard and very few picked up these two final marks, but this is a topic that should not be beyond students at this level.

Q19.

Most of the candidates had part (a) correct, with the majority choosing to substitute $t = \frac{3}{y}$ into the given equation for x rather than to substitute into the given cartesian equation. The commonest, but rarely seen, error was to not square the 3 or to make a sign error.

The majority of candidates chose to approach part (b)(i) through parametric differentiation, and most did it accurately, with many gaining full marks. The most common error was to lose the minus sign on the –16 gradient, having initially found this correctly. Others made mistakes with the fractions.

For part (b)(ii), there are numerous ways in which this intersection can be verified, and many different ways were attempted by the candidates. Many scored 1 mark by finding the y - value, either directly from their tangent or from the curve, or even via the parametric equations.

Some substituted their tangent equation into the curve and attempted to verify that $x = \frac{3}{2}$ satisfied the equation. Some did this convincingly, but others proceeded rather more in hope than expectation. Some tried to factorise the resulting cubic equation. The most efficient method was to find the y-value from the equation of the tangent and then verify that ($\frac{3}{2}$, –8) satisfies the equation of the curve. This method was mostly seen from those who achieved full marks on the question.

Q20.

In part (a)(i), most students were able to find the correct gradient, although some insisted on writing the gradient as $\frac{4}{3}x$. Unfortunately, quite a few were unable to make y the subject of the equation $4x - 3y = 7$ and, consequently, the incorrect value $-\frac{4}{3}$ was offered as the gradient by a significant number of students; a minority confused the gradient with the y-intercept and gave answers of $\pm\frac{7}{3}$.

In part (a)(ii), the most successful attempts at the equation of the line used an equation of the form $y - y_1 = m(x - x_1)$, as flagged above. Many who tried to use $y = mx + c$ lost marks because they made arithmetic errors when trying to find the value of c. Quite a large number did not appreciate that parallel lines have the same gradient and these usually found the negative reciprocal of their gradient from part (a)(i) to use for the gradient of the parallel line. Although most students found a correct equation, many mistakes were seen when trying to rearrange their equation into the required form with integer coefficients.

In part (b), most students made an attempt at the simultaneous equations, but it was alarming to see how few obtained the correct solution. Poor algebraic skills and carelessness were often evident here; elimination of one of the variables proved to be the most successful approach; those using substitution were usually unable to cope with the fractions and negative signs. Some students did not read the question carefully, and no credit was given to those who used the wrong pair of equations.

In part (c), once again, poor algebra was the main problem. Many of those students who successfully wrote $4(k - 2) - 3(2k - 3) = 7$ were unable to solve the equation for k correctly.
Others scored no marks for substituting into the wrong equation.

Q21.

In part (a), some weaker students did not seem familiar with the standard equation of a circle, with several writing the left hand side of the equation as $(x + 5)^2 + (y + 8)^2$ or $(5 - a)^2 + (8 - b)^2$. However, most errors were associated with the value of k; typical incorrect values seen on the right hand side of the equation were 5, 89 and $\sqrt{89}$.

In part (b)(i), the verification mark was rarely scored. Many, who had the correct circle equation, were able to verify that A was a point on the circle but then neglected to make a concluding statement. Those who had the wrong value of k in part (a) could not earn this mark unless they argued in terms of Pythagoras, stating the correct value of the radius.

In part (b)(ii), while most students found the gradient of AC correctly, many used the value of the gradient of AC when finding the equation of the tangent. It was pleasing to see a large number of students realise the need to find the negative reciprocal, but these students were sometimes unable to present their tangent equation in the required form.

In part (c)(i), most used the distance formula correctly and found that CM^2 was equal to 20 and then deduced that $CM = 2\sqrt{5}$. Occasionally the surd was not handled correctly, with some writing the final answer as $4\sqrt{5}$. No marks were awarded to students who simply wrote down the answer $2\sqrt{5}$ without supporting working.

Part (c)(ii) was not answered quite so well. Pythagoras was not always used correctly as the angle PCQ at the centre of the circle was often assumed to be a right angle or the length CM assumed to equal PM . Even those who found the correct value of PM either found the area of only half the triangle or were unable to deal with the product of the surds.

Q22.

In part (a)(i), the idea of y being increasing when $\frac{dy}{dx} > 0$ seemed unfamiliar to many students, who seemed content to substitute a few numerical values for x into the expression for $\frac{dy}{dx}$ rather than tackling the three-line proof. Consequently, very few scored full marks on this part.

Many students ignored the inequality signs throughout part (a)(ii). It was pleasing to see many students attempting to factorise the quadratic, often successfully, rather than relying on the formula. Perhaps more students would be successful in solving quadratic inequalities if they were to draw a sign diagram or sketch a graph. There were many correct solutions, but students must realise that $x < 2$, $x > \frac{4}{3}$ is not the same as $\frac{4}{3} < x < 2$

In part (b)(i), most students attempted to evaluate $\frac{dy}{dx}$ when $x = 2$. Not all were able to cope with the evaluation of $-6\,(2)^2$, which they needed to do in order to convince examiners that $\frac{dy}{dx}$ was indeed equal to zero. Once more, some failed to write a concluding statement or referred to a stationary point rather than the fact that the tangent was parallel to the x-axis.

In part (b)(ii), it was pleasing to see some fully correct solutions. Most recognised that the equation of the tangent at P was $y = 3$, but it was disappointing to see a number of students write $y - 3 = 9(x - 2)$ followed by $y = x + 1$ as their tangent equation. In finding the equation of the normal at Q, some, having correctly found that $\frac{dy}{dx} = 10$ when $x = 3$, used the value 10 instead of $-\frac{1}{10}$ for the gradient of the normal. Arithmetic errors were very common by those who used $y = mx + c$ when finding the value of c; typically $y = \frac{x}{10} + \frac{7}{10}$ was seen instead of $y = \frac{x}{10} - \frac{13}{10}$. Curiously, quite a few students who had previously obtained $y = 3$ as the equation of the tangent, and consequently the correct y-coordinate of the point R, then substituted the correct value of x into their incorrect equation for the normal and found a different, and now incorrect value for the y-coordinate of R.

Q23.

In part (a)(i) Although the majority of candidates found the correct gradient, some lost the – sign and others wrote their answer as $-\frac{3}{2}x$. Those who used two correct points often made a sign error in simplifying their expression for the gradient.

In part (a)(ii) it was disconcerting to see so many candidates finding the equation of a line perpendicular to AB rather than the equation of a line parallel to AB. Even having the correct line equation did not mean that full marks were earned; quite a few found the value of x when $y = 0$ and others who correctly obtained the equation $3x + 2y + 8 = 0$ quoted the intercept as being at $y = -8$.

Part (b) was generally well done, although eliminating y and getting as far as $5x = -5$ did not always result in the correct solution; final answers of $x = -5$ or $x = 1$ were seen far too often.

In part (c) the relevant distance formula expression often had a sign error in at least one bracket or was equated to 5 instead of 25. However it was good to see a sizeable number of fully correct solutions. Some candidates used a diagram and a vector approach and wrote down the correct values of k.

Q24.

In part (a)(i) candidates who recognised that the first step required was differentiation generally produced correct answers, but a few had trouble with the "$-x$" term and others were unable to combine $-1 + (-4)$ correctly. Some weaker candidates assumed that they could find the gradient using the coordinates of two points.

In part (a)(ii) once again there was much confusion between parallel and perpendicular lines, with many candidates giving the wrong gradient of the tan gent. To earn the final mark it was essential to express the equation in the specified form and some failed to do so; for instance, an answer such as $y + 5x = 17$ is not in the correct form.

In part (b)(i) the integration was generally correct and most candidates made an attempt to substitute the correct limits. A few however could not integrate “14” and left it as a constant. Very few were able to combine their fractions correctly and consequently only a small proportion of the candidates found the correct value of the integral.

In part (b)(ii) candidates who had only found the definite integral in part (b)(i) were generously rewarded with the remainder of the marks here. Not everyone recognised the need to find the area of a triangle in order to find the area of the shaded region. A few correctly found the area under the line $y = 4x + 8$ by integration, but this is not the recommended approach.

Q25.

In part (a)(i) the LHS of this equation was almost always correct but the RHS was less commonly so with $\sqrt{13}$ or 13^2 being common wrong answers.

In (a)(ii) again, the first 4 terms, namely $x^2 + y^2 + 6x - 4y$, were generally correct with just the odd slip occurring, either in a sign or perhaps from not doubling one of the x or y coefficients. The rest of the equation was only completed correctly by about half of the candidates, although some managed to recover from an error in their equation in part (a)(i), usually replacing $\sqrt{13}$ by 13.

Part (b) defeated many candidates. Using algebra, putting $x = 0$ and solving the quadratic to find the y-intercepts, and hence finding the length of AB, was done only by the most competent. However some candidates were successful using a diagram and the theorem of Pythagoras. Unfortunately some who drew a picture assumed that the two radii to the intercepts were at right angles and their Pythagoras calculations were incorrect.

Part (c)(i) proved an insurmountable hurdle for those who tried to use their wrong circle equation; fortunately those who used the form given in the question were more likely to be successful. Some compared the distance from the point to the centre of the circle with the radius, but, in many cases, there was some confusion between 13 and $\sqrt{13}$ and these candidates lost the mark. Unfortunately some also lost the mark by failing to complete their verification with a relevant statement.

Part (c)(ii) was the only place in this question paper where it was essential to know how to find the gradient of the line joining two points. A few candidates had their expression upside down; however many who wrote down a correct expression for the gradient were often unable to cope with the negative signs when simplifying their gradient into a simple fraction.

In (c)(iii) the tan gent to the circle required the negative reciprocal of the gradient of CD to be calculated, since a perpendicular line was involved, but many seemed unaware of this procedure.

Q26.

In part (a) most candidates were able to find the correct gradient; however, some were unable to make y the subject of the equation $7x + 3y = 13$.

In (b)(i) the most successful attempts at the equation of the line used an equation of the

form
$y - y_1 = m(x - x_1)$ as flagged above. Many who tried to use $y = mx + c$ lost a mark because they made arithmetic errors when trying to find the value of c. Unfortunately, some candidates used the gradient of the line perpendicular to AB.

In part (b)(ii) many candidates used a method based on differences of coordinates or simply wrote down the correct coordinates of A, but others misunderstood the request and actually found the mid-point of A and the given point.

Part (c) Most candidates made an attempt at the simultaneous equations but it was alarming to see how few obtained the correct solutions. Poor algebraic skills and carelessness were often evident here; elimination of one of the variables proved to be the most successful approach; those using substitution were usually unable to cope with the fractions. No credit was given to candidates who used the wrong pair of equations.

Q27.

In part (a) most candidates found the correct values of a and b, but correct values for k were not so common. Some sloppiness was again evident with candidates failing to write squared outside the brackets or omitting the plus sign between the terms on the left hand side.

In (b) some used an approach based on Pythagoras, but the majority of candidates used an algebraic method, substituting $y = 0$ into their circle equation and solving the resulting quadratic equation in x. Although there were a number who made sign errors, many candidates produced the correct values of x. Nevertheless, common mistakes such as putting $x = 0$ or $(y + 8) = 0$, instead of $y = 0$, resulted in no marks being earned.

In part (c) many candidates assumed that the gradient of CA was 2.5 instead of finding the negative reciprocal and many who did have a correct equation were unable to produce an equation with integer coefficients as requested. A sketch might have helped some to have seen the correct configuration and perpendicularity.

In (d)(i) most candidates realised the need to substitute $y = 2x + 1$ into one or other form of their circle equation but algebraic errors abounded as they tried to produce the displayed quadratic equation. Those who expanded the circle equation first were generally more successful; others who did not have 100 on the right hand side of their circle equation failed to recognise this error when they failed to produce the printed answer for the equation.
Candidates should be strongly discouraged from starting each new line with an equals sign when producing an algebraic proof of this type that involves equations.

In (d)(ii) those who completed the square failed to score full marks if they simply wrote $x + 3 = \sqrt{11}$ before concluding that $x = -3 \pm \sqrt{11}$. Those using the quadratic equation formula

often struggled to simplify $\dfrac{-6 \pm \sqrt{44}}{2}$.

Q28.

Many candidates did not have enough room to complete this question in the space available, often because of over–elaborating their answer to part (b)(ii).

Part (a) was answered correctly in the majority of cases.

Part (b)(i) was also done well, with only a small number making slips with signs, or square-rooting when they should have squared.
Part (b)(ii) had only 1 mark allocated and a request to “write down” the required equation. Although approximately 25% of the candidates did write down the correct equation, $x = 4$, of the normal at the maximum point, a far more popular but wrong answer was $y = 8$. Many other candidates produced half a page of working to reach an equation of a line which was neither vertical nor horizontal.

It was pleasing to see a much better response in part (c)(i) with many more finding the correct equation of the normal at P, although many failed to reduce it to a form with positive integers a, b

and, in particular, c. It was not uncommon to see the final answer left as ‘ $2x+3y=\frac{99}{4}$ ’, just one
step away from the correct answer ‘$8x + 12y = 99$’. In general, only the more able candidates, who made good use of the given diagram to show that the normal at M was vertical, gained credit for their answers to part (c)(ii) of the question.

Q29.

The candidates were mostly familiar with the techniques associated with logarithms, but the relationships between a and c, and between b and m, were often left in logarithmic form rather than in the way specified in the question, which involved exponential forms. These exponentials could perhaps have pointed candidates in the right direction when they came to answering part (c), where antilogarithms were needed for both answers. The linear graphs in part (b) were mostly plotted and drawn accurately, though some candidates misused the scale on the vertical axis.

Q30.

In part (a)(i), although some made arithmetic errors when finding the gradient of AB, the majority of answers were correct. It was necessary to reduce fractions such as 4/2 in order to score full marks. In part (a)(ii), those who chose to use Pythagoras’ Theorem, calculating lengths of sides to prove that the triangle was right angled, scored no marks here. The word “hence” indicated that the gradient of AB needed to be used in the proof that angle ABC was a right angle. A large number of those using gradients failed to score full marks on this part of the question. It was **not** sufficient to show that the gradient of BC was –1/2 and then to simply say “therefore ABC is a right angle”; an explanation that the product of the gradients was equal to –1 was required.

In part (b)(i), most candidates were able to find the correct coordinates of the mid point, although a few transposed the coordinates and others subtracted, rather than added, the coordinates before halving the results.

In part (b)(ii), it was rare to see a solution with all mathematical statements correct. Too often candidates wrote things like $AB = 2^2 + 4^2 = 20 = \sqrt{20}$ and, although this was not penalised on this occasion, examiners in the future might not be quite so generous. It was surprising how many candidates did not know the distance formula. Some wrote down vectors but, unless their lengths were calculated, no marks were scored.

In part (b)(iii), many candidates found an equation of the wrong line. The line of symmetry was actually BM, although some chose an equivalent method using the gradient of a line perpendicular to AC. The most successful candidates often used an equation of the form $y - y_1 = m(x - x_1)$; far too often those using $y = mx + c$ were unable to find the correct value of c, usually because of poor arithmetic.

Q31.

In part (a), the slightly unusual form of the initial quadratic caused problems to those who forgot to add 2 after multiplying out the brackets. The first term in the completion of the square was done successfully by most candidates, although the answer $(x - 4)^2 - 3$ was seen almost as often as the correct form. It was unfortunate that an error at this stage was seldom picked up by candidates when they went on to sketch the curve.

In part (b)(i), the sketch was usually of the correct shape but many drew a parabola with the vertex in the wrong quadrant, or the y-intercept was incorrect. The question did ask for the coordinates of the minimum point and this was not always stated by candidates.

In part (b)(ii), a few candidates immediately wrote down the correct equation for the tangent, whereas many felt the need to differentiate in order to find the gradient of the curve at the vertex. Many candidates seemed unaware that the vertex was actually the minimum point and that the tangent at the vertex would be parallel to the line $y = 0$.

In part (c), the more able candidates earned full marks. The term **translation** was required and, although there seemed to be an improvement in candidates' using the correct word, it was still common to see words such as "shift" or "move" being used instead. Others thought that simply writing down a vector was enough. A very commonly stated, but incorrect, vector was $\begin{bmatrix} -4 \\ 1 \end{bmatrix}$.

Q32.

In part (a)(i), many candidates did not realise that differentiation was required in order to find the gradient of the curve, but instead erroneously used the coordinates of O and A. Some tried to rearrange the terms but usually made sign errors in doing so. In part (a)(ii), those who had the correct gradient in part (a)(i) were usually successful in finding the correct equation of the normal, though not everyone followed through to the required form, and sign errors were common. However, most obtained at least a method mark here, unless they found the equation of the tangent. The main casualties were once again those who always use the $y = mx + c$ form for the equation of a straight line.

In part (b)(i), most candidates were well drilled in integration and scored full marks, although some wrote down terms with incorrect denominators. The limits 2 and 0 were usually substituted correctly, but it was incredible how many could not evaluate 32 – 38 – 8 without a calculator. The correct value of the integral was –14, but far too many thought that they had to change their answer to +14 and so lost a mark. A small number of candidates differentiated or substituted into the expression for y rather than the integrated function.

In part (b)(ii), there was still some apparent confusion about area when a region lies below the x-axis. The area of the triangle was 6 units and hence the area of the shaded region was 14 – 6 = 8 but, not surprisingly, there were all kinds of combinations of positive and negative quantities seen here. It was worrying to see so many candidates failing to calculate the triangle area, with several finding the length of OA instead. Some able candidates found the equation of OA and the area under it by integration, but this was not the expected method.

Q33.

Part (a) Many candidates were unable to make y the subject of the equation $2x + 3y = 14$ and, as a result, many incorrect answers for the gradient were seen. Those who tried to use two points on the line to find the gradient were rarely successful.

Part (b)(i) Those candidates who obtained a value for the gradient in part (a) were usually aware that the line DC had the same gradient. Those using $y = mx + c$ often made errors when finding the value of c, whereas those writing down an equation of the form $y - y_1 = m(x - x_1)$ usually scored full marks.

Part (b)(ii) Most candidates realised that the product of the gradients of perpendicular lines should be –1 and credit was given for using this result together with their answer from part (a). Careless arithmetic prevented many from obtaining the final equation in the given form with integer coefficients.

Part (c) Although many correct answers for the coordinates of B were seen, the simultaneous equations defeated a large number of candidates. No credit was given for mistakenly using their equation from part (b)(i) or part (b)(ii) instead of the correct equation for AB, clearly printed below the diagram. Many did not recognize the need to use the equation of AB at all. It was common to see $x = 0$ or $y = 0$ substituted into the equation for BC and then solved to obtain the other coordinate.

Q34.

Part (a) Most candidates obtained the correct left hand side of the circle equation but many failed to recognize that the radius was 5. Alarmingly many thought that r was equal to –5 or wrote down the right hand side of the equation as -5^2, thus displaying a fundamental misunderstanding of the idea of radius as a length.

Part (b)(i) Most who had the correct circle equation were able to verify that the circle passed through the point P, although those who neglected to make a statement as a conclusion to their calculation failed to earn this mark.

Part (b)(ii) The negative signs caused problems for many when finding the gradient of PC and only the better candidates obtained the correct value. Many candidates then found the negative reciprocal of this fraction instead of using the gradient of PC to find the normal to the circle at the point P.

Part (b)(iii) There were basically two approaches to this question, although some candidates were merely guessing and no credit was given for a correct answer without supporting working.

The most common method involved distances or squares of distances; many made errors in finding the coordinates of M and then struggled with the fractions when squaring and adding to find the length of PM; whereas others noted that the length of PM was simply half the radius. A simple comparison with the length of PO led to the correct conclusion.

The second approach was essentially one using vectors or the differences of coordinates, but this method was not always explained correctly and left examiners in some doubt as to whether candidates really understood what they were doing. The best candidates wrote down the correct vectors $\overrightarrow{PM}$ and $\overrightarrow{OP}$ and reasoned that these vectors had the same y-component but different x-components and it was then easy to deduce that P was closer to the point M.

Q35.

In part (a) most candidates were able to find the correct coordinates of the mid point, although a few transposed the coordinates and others subtracted rather than adding the coordinates before halving the results.

Full marks were only awarded in part (b) for a gradient of –2 and quite a few candidates did not give their answer in this simplest form.

In part (c)(i) most candidates realised that the product of the gradients should be –1. However, not all were able to calculate the negative reciprocal. Others used an incorrect point such as A or B and therefore found an equation of the wrong line. The most successful used an equation of the form $y - y_1 = m(x - x_1)$ as flagged above. The printed answer helped most candidates to be successful in finding the correct equation of the line.

In part (c)(ii) most candidates made an attempt at this part of the question, but the failure to use brackets for the second term caused the majority to find an incorrect value for k. Others foolishly tried to substitute $x = k$ and $y = k + 5$ into their own incorrect line equation rather than using the printed answer from part (c)(i).